

1.3 Types of User Interaction

تعامل کی اقسام میں براہ راست ہیرا پھیری، کمانڈ، مینو، فارم، گفتگو (چیٹ بوٹ)، اشارہ، آواز، سیاق و سباق سے آگاہ...

CS XII

Chapter 1: Computer Systems (HCI)

Lecture 05 of 15

1.3 Types of User Interaction

LEARN IT

User interaction can happen in many different ways depending on the technology used. Learn every type with one clean example — board questions often ask you to "list and explain" five or more.

Interaction type	Description	Examples
Direct manipulation	Handling digital objects on a screen exactly like physical objects.	Dragging files, zooming images with fingers
Command-based	Giving precise instructions through typed text commands.	MS-DOS, Linux terminal (<code>dir</code> , <code>cd ..</code>)
Menu-driven	Selecting from predefined lists or options instead of typing.	Clicking "Start Test" or "Download" on a portal
Form fill	Entering structured data into blank text fields.	Sign-up forms, ticket booking screens
Conversational	Communicating with systems using natural human language (text or voice).	Chatbots, ChatGPT, Google Assistant
Gesture-based	Using body or hand movements to control systems.	Swiping touchscreens, VR hand gestures
Voice interaction	Speaking commands directly to the system.	"Call Aslam" on a smartphone
Context-aware	Systems sense the environment and act automatically without a prompt.	Smart lighting turning on when you enter a room
Biometric	Using biological features for interaction and authentication.	Fingerprint unlock, FaceID
Haptic and tactile	Providing physical feedback to confirm actions.	Phone vibration when typing
Multimodal	Combining two or more interaction types together.	Voice + touch (speak a search, then tap)

WORKED EXAMPLE

Worked example: a full JazzCash "Send Money" session, labelled by type.

1. Open the app with a fingerprint → **biometric**.
2. Home screen shows Send Money, Bill Payment, Load → you tap Send Money → **menu-driven**.
3. Type the receiver's mobile number and the amount → **form fill**.
4. The keypad vibrates on each tap → **haptic**.
5. Optional: say "Send five hundred to Ali" then tap Confirm → **multimodal** (voice + touch).

Same idea on a **school portal**: menu-driven ("Date Sheet", "Fee Challan"), form fill (leave application), conversational (a chatbot "When is the chemistry practical?").

REMEMBER IT

Group them: hands on screen (direct, gesture, haptic) · words (command, conversational, voice) · lists and blanks (menu, form) · automatic/body (context-aware, biometric) · mix (multimodal).

COMMON MISTAKE

Mistake: Treating "voice interaction" and "conversational" as identical, or calling a dropdown menu "command-based".

Fix: Voice is spoken commands only. Conversational includes typed chat (ChatGPT) as well as voice. Command-based means you type precise codes such as `dir`, not clicking a menu.

اردو میں سمجھیں

تعامل کی اقسام میں براہ راست ہیرا پھیری، کمانڈ، مینو، فارم، گفتگو (چیٹ بوٹ)، اشارہ، آواز، سیاق و سباق سے آگاہ نظام، بایومیٹرک، بیپٹک اور ملٹی موڈل شامل ہیں۔ جاز کیش میں فنگر پرنٹ بایومیٹرک ہے، "رقم بھیجیں" مینو ہے، نمبر لکھنا فارم ہے، اور آواز کے ساتھ ٹیپ ملٹی موڈل ہے۔

CHECK YOURSELF

1. Which type is a Linux `cd ..` command?
2. Smart lighting that turns on when you enter a room is which type?
3. Why is speaking a search and then tapping a result called multimodal?

Answers: 1. Command-based. 2. Context-aware. 3. Because it combines two or more types (voice + touch) in one task.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. GUI means:

- a. Graphical User Interface
- b. Great Universal Internet
- c. Gate Unary Inverter
- d. Grouped Unique Index

MCQ 2. CLI is:

- a. Command-line (typed commands)
- b. Only touch
- c. Only VR
- d. Only pie menus of 1990 Mac always

MCQ 3. Voice UI example:

- a. Siri / Alexa / Google Assistant
- b. A datasheet
- c. A K-map
- d. A foreign key

MCQ 4. Gesture UI example:

- a. Swipe, pinch, Wii/Kinect-style move
- b. SQL SELECT
- c. Bubble swap
- d. HTTP GET only

MCQ 5. The best interaction type depends on:

- a. User, task, environment
- b. Whatever is newest
- c. Always voice
- d. Always CLI

MCQ 6. A surgeon in a sterile field may prefer:

- a. Voice/gesture over touch
- b. A dusty mouse
- c. Tiny 4pt CLI
- d. Captcha on the pump

Extra short questions

1. Name four interaction types with examples.
2. One strength of CLI.
3. One weakness of voice.
4. Can one app offer two types?

Extra long question

1. Recommend an interaction type for: (a) programmer (b) ATM (c) car while driving (d) museum kiosk. Justify with user+environment.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. GUI desktop; CLI terminal; touch; voice; gesture; AR/VR — any four.
2. Fast and scriptable for experts.
3. Noise, privacy, accents, no visual scan of options.
4. Yes — VS Code GUI + terminal; phones GUI + voice.

Long

1. (a) GUI+CLI (b) GUI+keys, privacy (c) voice/steering-wheel, eyes up (d) large touch, language choice. Table.